



MAPÚA MALAYAN COLLEGES MINDANAO

**SYSTEMS ANALYSIS AND DESIGN
PART I – SYSTEMS ANALYSIS REPORT**

Amigo School of Calinan, Inc.

**EduGuide PH: An AI-Powered Study
and Career Guidance System**

Submitted in Partial Fulfillment of the Requirements
for the Course in Systems Analysis and Design

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I. INTRODUCTION

The fast development of artificial intelligence (AI) and information and communication technology (ICT) has revolutionized educational systems around the world. In the Philippines, digital learning technologies are becoming increasingly important, particularly in places with limited access to tutors, learning materials, and career coaching. According to the Department of Education (DepEd), many students in rural and semi-urban areas struggle with lesson comprehension and career preparation due to a lack of individualized academic support.

Amigo School of Calinan, Inc., a private K-12 school in Calinan, Davao City, now uses traditional classroom education and manual guidance counseling. While the school is dedicated to providing high-quality education, limited faculty and counseling resources make it difficult to provide personalized learning support and career advising to all students.

EduGuide PH is a web-based AI study and career coach that enables students to receive individualized learning support. The system includes an AI chatbot that can respond to academic questions, explain lessons, and recommend career paths based on user input. It supports three languages (English, Tagalog, and Cebuano) and is designed for mobile devices, making it perfect for students in places with restricted internet access. This Systems Analysis Report assesses the current academic support and career guidance processes at Amigo School of Calinan, identifies problems and opportunities, defines system requirements, and recommends using EduGuide PH as a technological solution to improve student learning outcomes and career readiness.

II. THE ORGANIZATION

Company Profile and History

Amigo School of Calinan, Inc. is a private educational institution located on Dr. de Lara Street Calinan, Davao City. The school, which is registered as a business, provides basic education from Kindergarten through Grade 12. Its purpose is to provide accessible and high-quality education to learners in the Calinan neighborhood and adjacent areas.

The school is still in the early stages of facility and technology development. While it maintains a viable learning environment, use of digital tools and automated systems is limited. The school's administration acknowledges the need of using technology to improve both teaching and learning results.

Organizational Structure

The school has a standard private school organizational hierarchy, with the School Director at the front, followed by the Principal, Academic Coordinators, Grade Level Advisers, Subject Teachers, and the Guidance Office. Mary Grace Nakano, the Guidance Associate, is the primary point of contact for student and career counseling.

Workforce

The Amigo School of Calinan employs about 20-30 teaching and non-teaching people, including administrators, faculty members, administrative staff, and guiding personnel. The student population ranges from kindergarten to grade 12.

Existing ICT Infrastructure

Currently, the school relies on basic ICT resources such as desktop PCs in the faculty office and a computer lab with restricted internet access. The guidance office uses spreadsheet software to manage student records. There is no centralized digital infrastructure for academic support, tutoring, or career guidance. Students generally rely on texts, instructor consultations, and peer support to help them learn.

III. VISION AND MISSION

Vision

ASCI dreams of its students who passionately love their country and whose values and competencies enable them to realize their full potential and contribute meaningfully to building the nation.

Mission

ASCI protects and promotes the right of every student to quality, equitable, culture-based, and complete basic education where students learn in a child-friendly, gender-sensitive, safe, and motivating environment. Teachers facilitate learning and constantly nurture every learner.

IV. BUSINESS ENVIRONMENT

Amigo School of Calinan provides private basic education. The school competes with other private and public schools in the Davao City area. The key environmental influences are:

- **Rising demand for digital learning** - Parents and students are increasingly expecting technological integration in schools.
- **Limited access to tutors** - Many students are unable to afford private tutoring, leaving a gap in customized academic help.
- **Career guidance needs** - Students seek aid in finding career paths that are compatible with their interests and academic achievement.
- **Government initiatives** - The Department of Education encourages the use of ICT in education through initiatives such as DepEd Commons and computerization projects.

V. CRITICAL SUCCESS FACTORS

No.	Critical Success Factor	Description	Impact on Operations
1	Personalized academic support	Students receive help with difficult lessons outside class hours	Improves learning outcomes and reduces failure rates

2	Accessible learning resources	Study materials are available anytime, even on mobile devices	Increases student engagement and independent learning
3	Career guidance availability	Students can explore career options based on their interests	Enhances career readiness and decision-making
4	Teacher workload reduction	Automated assistance reduces repetitive student queries	Allows teachers to focus on higher-value instruction
5	Inclusive language support	Students can interact in Filipino, Cebuano, or English	Broadens accessibility for diverse learners
6	Performance tracking	Monitoring of student queries and quiz results	Enables data-driven interventions

VI. THE CURRENT SYSTEM

Overview

Amigo School of Calinan presently provides academic support and career guidance through both manual and face-to-face approaches. Students who require extra assistance must approach teachers before or after class. Career assistance is provided through regular counseling appointments with the assistance Associate.

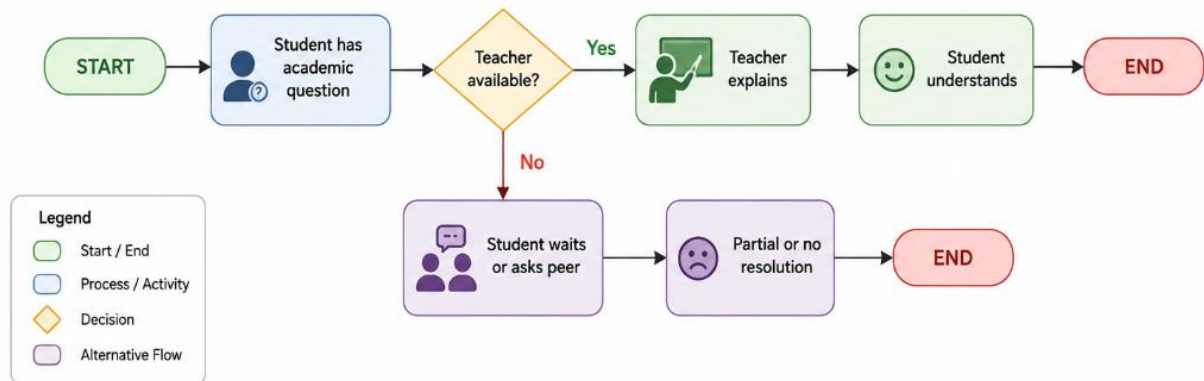
Academic Support Process

When a student has a question about a lesson, they usually wait until the break or after class to ask the teacher. If the teacher is unavailable, the student can ask peers or consult texts. There is no after-hours assistance mechanism.

Career Guidance Process

Students interested in career planning should schedule an appointment with the Guidance Associate. During the session, the student completes paper-based interest inventories while the counselor provides manual suggestions based on available facts. This process is time-consuming and subject to counselor availability.

Workflow Diagrams



VII. DESCRIPTION OF OPERATION

1. Academic Query Handling

Students ask inquiries in class or after school. Teachers respond in class or during consultation hours. Many questions remain unresolved because of time limits. There is no evidence of common issues.

2. Career Counseling

The Guidance Associate leads one-on-one sessions. Career assessments are administered on paper. Students receive generic counsel but little direct examination of job options.

3. Learning Resource Access

Students depend on printed modules and textbooks. There is no digital repository for explanations or extra resources.

VIII. PROBLEM / OPPORTUNITY DEFINITION (PIECES)

Category	Problem / Issue	Cause	Opportunity / Solution Direction
Performance	Slow response to student queries	Teacher availability limited to class hours	AI chatbot provides instant 24/7 responses
Information	No record of common student difficulties	No centralized query logging	System tracks questions and generates reports for teachers

Economics	High cost of private tutoring	Lack of affordable alternatives	Free AI-powered assistance reduces need for paid tutors
Control	No monitoring of student learning progress	Manual tracking only	Dashboard for teachers to view student activity
Efficiency	Counselor time spent on repetitive career advice	Manual assessments and consultations	Automated career suggestion engine reduces counselor workload
Service	Students lack after-hours support	No system available outside school hours	Web-based platform accessible anytime from mobile devices

IX. FUNCTIONAL REQUIREMENTS

ID	Requirement	User Role	Input	Output	Validation Rules
FR-01	User Login and Access Control	Student, Teacher, Admin	Username, Password	Authenticated session; role-based dashboard	Credentials must match database

FR-02	AI Chatbot for Academic Questions	Student	Text question (English, Tagalog, Cebuano)	AI-generated explanation	Question must be non-empty
FR-03	Career Path Suggestions	Student	Interests, grade level, strengths	List of suggested careers with descriptions	Input must be valid
FR-04	Suggested Prompts	Student	None	List of common academic topics	Display predefined list
FR-05	Voice Input	Student	Audio recording	Converted text query	Microphone permission required
FR-06	File Attachment	Student	Image or document	AI-assisted response based on attachment	File size $\leq$ 5MB

FR-07	Multilingual Support	Student	Language selection	Responses in chosen language	Supports English, Tagalog, Cebuano
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X. USE CASES

Actors:

- **Student** – primary user who asks questions and takes quizzes
- **Teacher** – views student progress and common difficulties
- **Admin/Guidance Counselor** – manages system, views career guidance analytics

Use Case Specifications

ID	Use Case	Actor(s)	Precondition	Main Flow	Postcondition
UC-01	Ask Academic Question	Student	Student logged in	Student types question → AI processes → Response displayed	Answer logged in history
UC-02	Receive Career Suggestion	Student	Student logged in	Inputs interests → AI suggests	Suggested careers saved

				careers → Student reviews	
UC-03	Register Account	Student	None	Provides name, grade, email → System validates → Account created	Student can log in

XI. FEASIBLE ALTERNATIVES

Alternative	Advantages	Disadvantages	Estimated Cost	Timeline
Alt 1 – Continue Manual System	No cost; familiar process	Limited support; no after-hours help; high counselor workload	0	N/A
Alt 2 – Basic Website with FAQs	Low cost; accessible	No interactivity; no	₱5,000 – ₱10,000	1–2 months

		personalization; no AI		
Alt 3 – EduGuide PH AI System	24/7 AI support; multilingual; career guidance; scalable	Higher initial cost; requires training and internet	₱30,000 – ₱60,000	3–4 months

XII. FEASIBILITY ANALYSIS

Feasibility Dimension	Weight	Alt 1 – Manual	Alt 2 – Basic Website	Alt 3 – EduGuide PH
Technical (20%)	0.2	3 (0.60)	4 (0.80)	4 (0.80)
Economic (25%)	0.25	5 (1.25)	4 (1.00)	3 (0.75)
Operational (20%)	0.2	4 (0.80)	4 (0.80)	4 (0.80)
Schedule (20%)	0.2	5 (1.00)	4 (0.80)	3 (0.60)
Legal/Security (15%)	0.15	2 (0.30)	3 (0.45)	5 (0.75)

Total Weighted				
Score	1	3.95	3.85	3.7

Ranking and Recommendations

- **Rank 1:** Continue the Manual System (short-term only)
- **Rank 2:** Basic website with FAQs.
- **Rank 3:** EduGuide PH AI System - Suitable for long-term strategic value.

While the manual method has the best short-term viability, EduGuide PH is suggested because it solves the core issues of poor academic support, lack of career advising, and student involvement. Its AI capabilities, multilingual assistance, and mobile accessibility make it the best solution for Amigo School in Calinan.

XIII. SCOPE AND LIMITATIONS

Scope

The proposed EduGuide PH system will cover:

1. AI-powered chatbot for academic questions
2. Career path suggestions based on student inputs
3. Multilingual support (English, Tagalog, Cebuano)
4. Voice input and file attachment features
5. Suggested prompts for common topics
6. User authentication

Limitations

1. Does not replace classroom instruction or teacher interaction
2. Requires internet connectivity for full functionality
3. No offline mode in initial deployment
4. Limited to text and image-based file attachments (no video processing)
5. Career suggestions are AI-generated and should be validated by counselors
6. Not integrated with DepEd's official learning management system
7. Student data privacy compliance requires additional configuration
8. Voice input accuracy depends on microphone quality and background noise

XIV. The Proposed System

The suggested solution is EduGuide PH, a web-based AI study and career mentor created for Philippine students. The system will be installed at Amigo School in Calinan to give 24-hour academic support, practice quizzes, and career guidance.

Technology Stack

Frontend: React

Backend: Node.js using Supabase

AI integration: Google Gemini API

Database: Supabase PostgreSQL

Authentication: Supabase Authentication

Hosting: Vercel

System Architecture

Students can access the platform via a web browser on their smartphones, tablets, or computers. The frontend routes user queries to the backend, which uses the AI API to generate responses. All user interactions, and career preferences are saved in the database. Teachers can visit a dashboard to view aggregated student queries.

Deployment Plan

- Phase 1: Local deployment on the school server for pilot testing with specific grade levels.
- Phase 2: Cloud hosting for increased accessibility and remote learning support.
- Phase 3: Integration with the school's current computer lab and guidance office procedures.

XV. GENERAL AND SPECIFIC OBJECTIVES

General Objective

To develop and implement an AI-powered study and career advice system (EduGuide PH) that improves academic support, learning outcomes, and career advising for students at Amigo School of Calinan.

Specific Objectives

- To provide students with immediate, 24-hour access to academic explanations via an AI chatbot.
- To enable self-paced learning using AI-generated quizzes and practice assessments.
- To provide tailored career route recommendations based on students' interests and academic strengths.
- To promote inclusive education through multilingual interactions (English, Tagalog, and Cebuano).
- To reduce teacher and counselor effort by automating responses to typical questions.
- To offer teachers with a dashboard that displays common student challenges for targeted education.
- To provide mobile accessibility, students using smartphones can use the system despite low internet connectivity.
- To set up a secure and privacy-compliant system for managing student data.

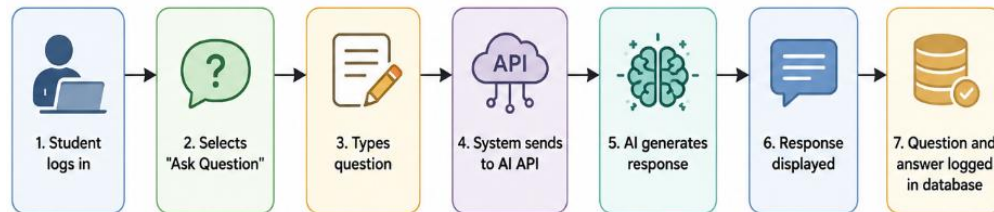
XVI. SCOPE OF AUTOMATION

The EduGuide PH system will automate the following processes at the Amigo School of Calinan:

1. **Academic query response** - Instead of waiting for an instructor, students receive rapid AI-generated responses.
2. **Career guidance intake** - Basic career choices are created automatically based on student inputs, saving counselor time during initial assessments.

3. **Student activity logging** – All queries, and career explorations are logged for teacher and counselor review.

Automation Flow



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